

Ch. 15 — P02 (C) Critical Analysis of a Published Case Study

Problem Bank | Chapter 15 | Type: C | Difficulty:

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Background

Reading simulation case studies critically is a core professional skill. A published study may have excellent methodology but poor communication, or vice versa. The Sargent V&V framework gives a structured lens for critique.

Case Study Summary (Fictional)

“Optimizing Ambulance Deployment in a Mid-Sized City”

Study objective: Determine the number and location of ambulances to minimise mean response time to 911 calls.

Model: A discrete-event simulation model of the city’s EMS system was built in a commercial package. Call arrivals were modelled as a homogeneous Poisson process at rate 12 calls/hr. Travel times were drawn from a uniform distribution [3, 15] minutes based on “analyst experience.” Ambulance service times (on-scene + transport + turnaround) were fixed at 45 minutes.

Validation: The model was “run until results stabilised” (no warmup analysis reported). Output was compared to the vendor’s benchmark database for “similar cities” — not to actual city data.

Results: With 6 ambulances, mean response time was 8.2 minutes (no confidence interval reported). With 8 ambulances, mean response time was 5.7 minutes. The study recommends 8 ambulances for a \$1.2M annual cost increase.

Limitations section: None.

Questions

- (a) **Conceptual validity issues:** Identify at least two assumptions in the conceptual model that are likely to be violated in a real EMS system. For each, explain the likely direction of bias (does it make performance look better or worse than reality?).

- (b) **Verification gaps:** What verification evidence is missing from the report? List at least three checks that should have been performed and described. For each, state what a failure would have indicated.
- (c) **Operational validity:** The validation used a vendor benchmark database rather than local historical data.
- Why is this a weaker form of validation than historical comparison?
 - Under what circumstances (if any) would benchmark comparison be acceptable as the primary validation evidence?
- (d) **Output analysis critique:** The report gives point estimates (8.2 min, 5.7 min) without confidence intervals.
- Construct a hypothetical scenario where the decision to add 2 ambulances would reverse if proper CIs were included.
 - What information (standard deviations, number of replications) would you need to test whether the difference is statistically significant?
- (e) **Limitations section:** Write a 3–5 bullet limitations section that should have appeared in this report. Follow the format: state the limitation, its likely magnitude of impact, and the condition under which it matters most.
- (f) **Recommendation:** Given the methodological gaps, would you use this study to justify the \$1.2M ambulance purchase? State your position clearly and explain what additional analysis you would require before making a recommendation to the city council.